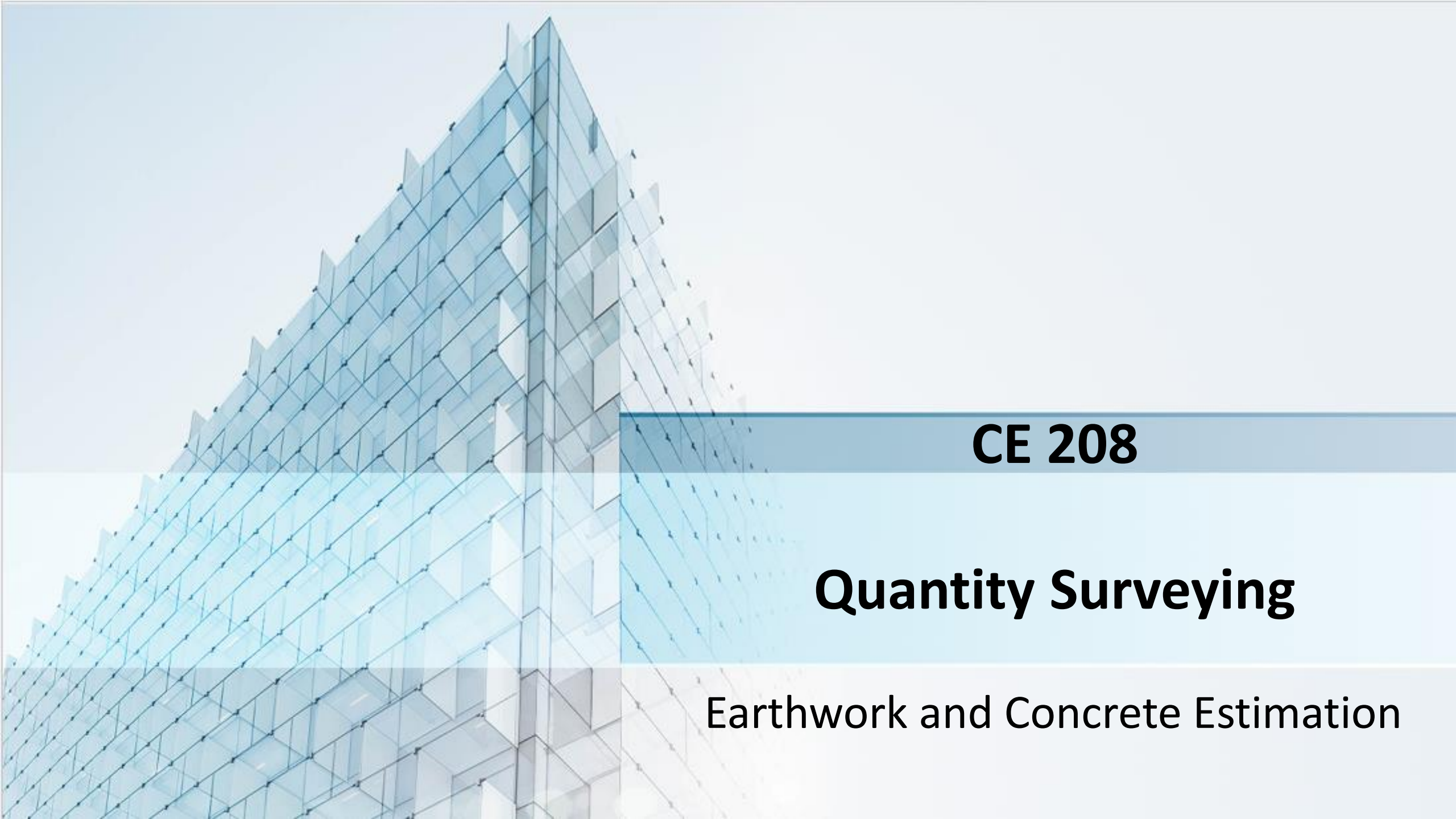


CE 208

Quantity Surveying

Earthwork and Concrete Estimation

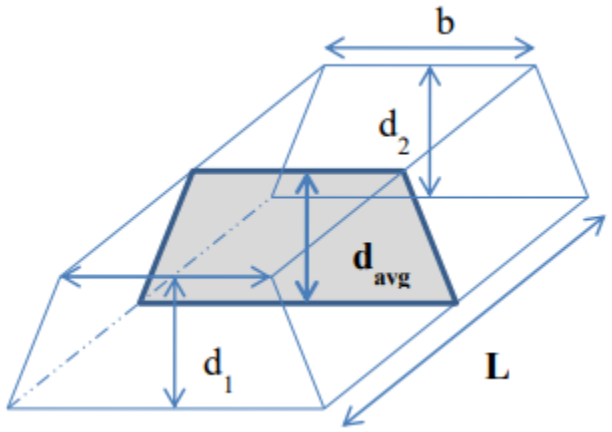
Calculation of Earthwork Volume

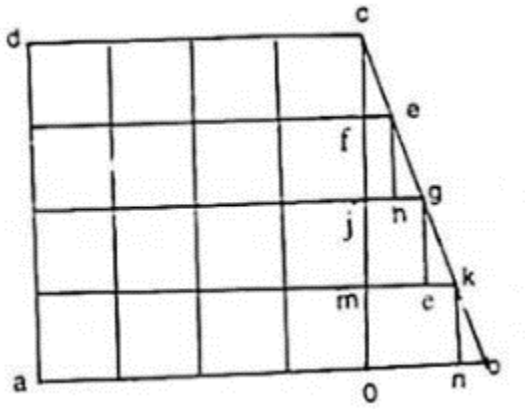
There are mainly three methods adopted-

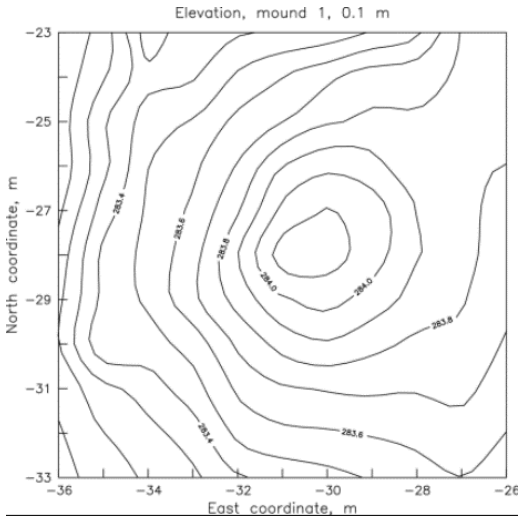
1. From Cross Sections: Measurement from cross section is a universally applicable method.

2. From Spot Levels: Measurement from spot levels are applied sometime for large excavation.

3. From contours: Rough estimates of volume may be made by treatment of the contour line and not much used in practice







Measurement from Cross- Section

- The total volume to be calculated is divided into a series of solids. The fundamental solids are prism, wedge and prismoid.
- The spacing of the sections depend on the character of the ground and accuracy required.
- The area of cross-section along the line are calculated by standard formulae and the volumes of prismoid betweent he cross-sections taken are then calculated by the following methods.

    1) Formulae of Mid-section method/ Average height method.

    2) Formulae of Trapezoidal method/ Average end area method/ mean-sectional area method.

    3) Formulae of Prismoidal method according to Simpson’s one-third rule.

Terms and Abbreviations

Station: A point where the level rod is held and not where level is set up. The distance between two consecutive stations depends on the topography of ground surface. 20m~50m for for plain country and 10m~20m for hilly country.

Roadway or Formation Width: It is the total width of the road including the carriageway and the two shoulders on the sides. That is,

Roadway Width= Carriageway Width + Shoulder Width

Terms and Abbreviations

EGL (Existing Ground Level) or GL (Ground Level): The existing earth surface.

FL (Formation Level): The proposed level of roadway.

RL (Reduced Level): A level stated in relation to a known bench mark or datum.

True Level: a level stated in relation to a Mean Sea Level (MSL).

- Cutting is required when EGL>FL

Longitudinal Slope/ Gradient: Gradient may be defined as the rate of rise or fall along the length of highway. It depends on the nature of traffic, nature of country and type of surfacing material. An optimum longitudinal level is where there is a balance of cut and fill to eliminate rolling nature of land.

Side Slope: Side slope is defined as the rate of rise or fall of the shoulders of the pavement. It depends on the soil characteristics and geographic location of the highway. For interstate systems they should not be steeper than 2 to 1. For fills less than 10' in height they are set at 4 to 1 or flatter. Always measured vertical: horizontal. Steep slopes create accident hazards.

- The height of banking and depth of cutting are designed considering the highest flood level, ruling gradient and to achieve maximum economy on earthwork.

Mid Section Formulae/ Average Height Method

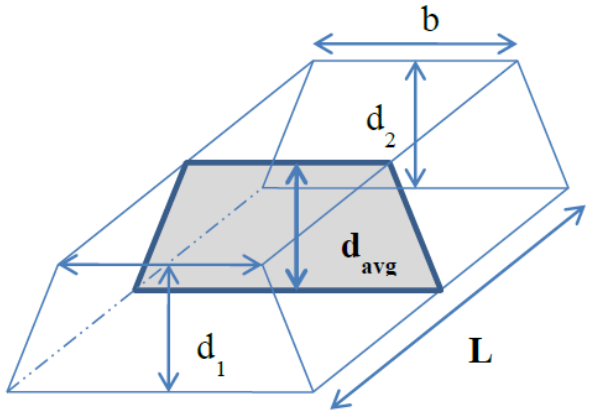
Depth section (1) = d_1 (note that d1 is the difference between GL & FL)

Depth section (2) = d_2 (note that d2 is the difference between GL & FL)

Average depth, d_{avg}= (d_1 + d_2)/2

Width of section = b

Side slope = 1: s (vertical: horizontal)



Area of mid-section, $A_{mid} = bd_{avg} + \left(\frac{1}{2}\right) sd_{avg}{}^2 + \left(\frac{1}{2}\right) sd_{avg}{}^2$

A_{mid}= (b+sd_{avg}) x d_{avg}

Length between two consecutive sections (between section (1) & (2)) = L_{1-2}

Volume of earthwork between these two consecutive sections (between section (1) & (2)), V_{1-2} = A_{avg} x L

V_{1-2} = (b+sd_{avg}) x d_{avg} x L (may be cut or fill)

Trapezoidal Formulae/ Average End Area Method/ Mean-Sectional Area Method

Depth section (1) = d_1 (note that d1 is the difference between GL & FL)

Depth section (2) = d_2 (note that d2 is the difference between GL & FL)

Area at end 1, A_1 = (b+sd_1) x d_1

Area at end 2, A_2 = (b+sd_2) x d_2

Mean sectional area, A_{mean}= (A_1+A_2)/2

Width of section = b

Side slope = 1: s (vertical: horizontal)

Length between two consecutive sections (between section (1) & (2)) = L

Volume of earthwork between these two consecutive sections (between section (1) & (2)), V_{1-2}=A_{mean} x L

$\boldsymbol{V_{1-2}}$ = A_{mean} x L (may be cut or fill)

Prismoidal Formulae

Depth section (1) = d_1 (note that d_1 is the difference between GL & FL)

Depth section (2) = d_2 (note that d_2 is the difference between GL & FL)

Area at end 1, A_1 = (b+sd_1) x d_1

Area at end 2, A_2 = (b+sd_2) x d_2

Mean sectional area, A mean = (A_1+A_2)/2

Width of section = b

Side slope = 1: s (vertical: horizontal)

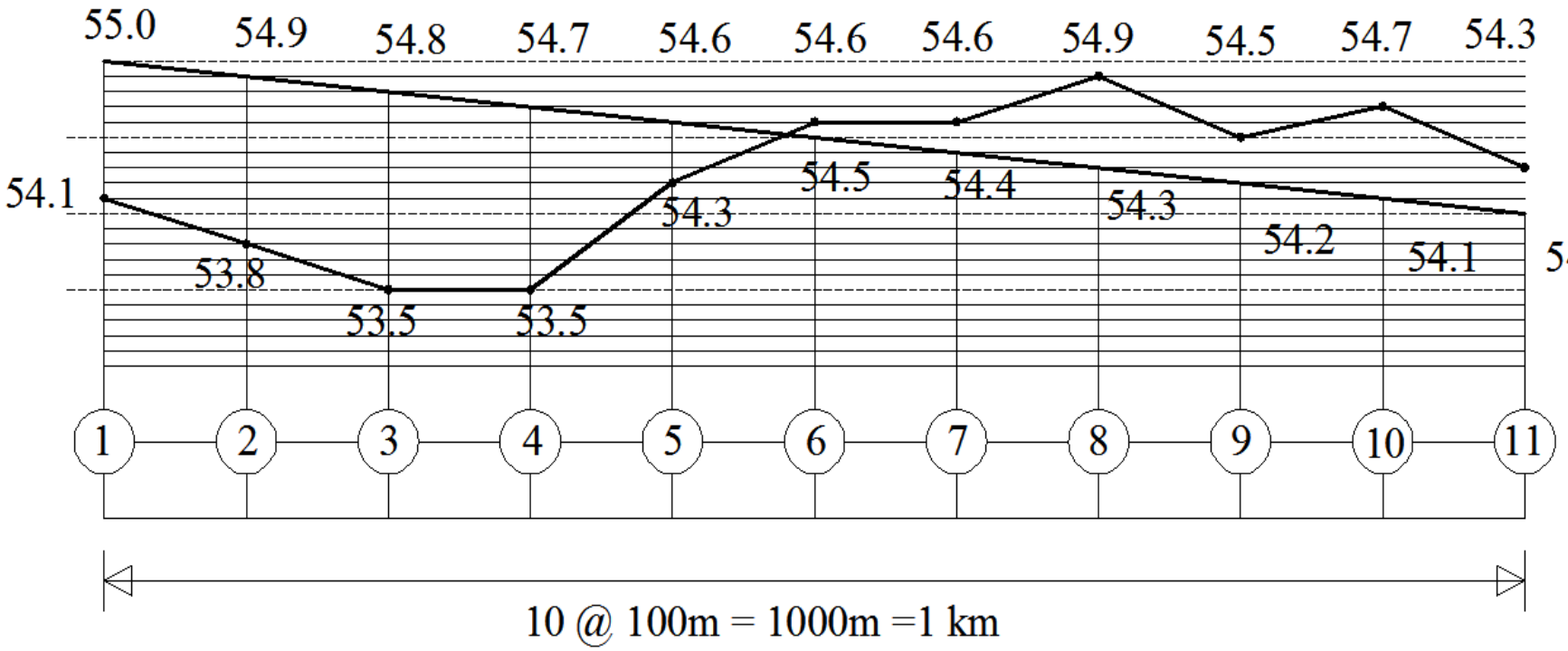
Length between two consecutive sections (between section (1) & (2)) = L

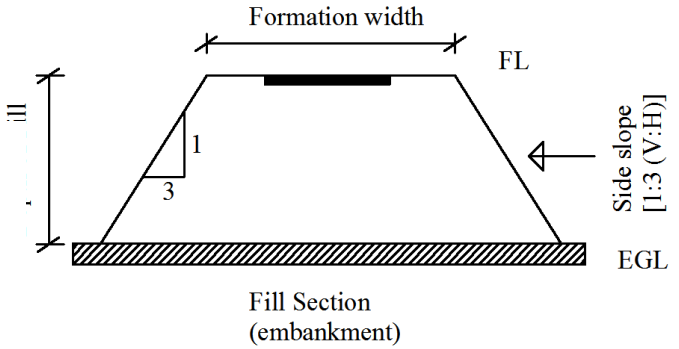
Volume of earthwork between these two consecutive sections (between section (1) & (2)),

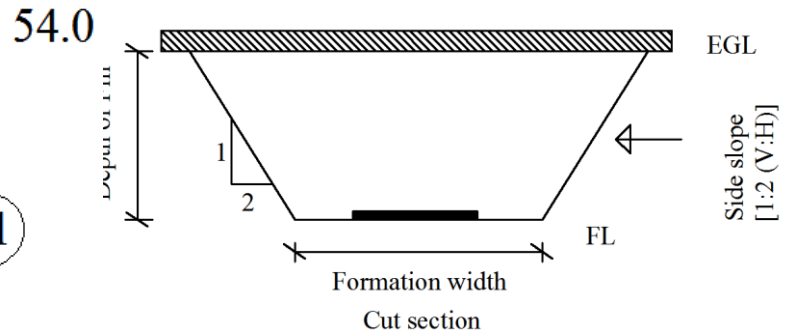
V_{1-2} = (A_1+ 4A_m+ A_2)/6 x L (may be cut or fill)

Worked out problem:

A 1 km road is to be constructed in existing ground level having reduced levels 54.1, 53.8, 53.5, 53.5, 54.3, 54.6, 54.9, 54.5, 54.7 and 54.3 meters at 100 m intervals. A required reduced level at station 1 is 55 meter and the downward gradient is 1 in 1000. The width of the road at formation level is 8 meter. Slopes to be maintained at cutting and filling are 1:2 (V: H) and 1:3 (V: H) respectively. Calculate the volume of Earthwork.

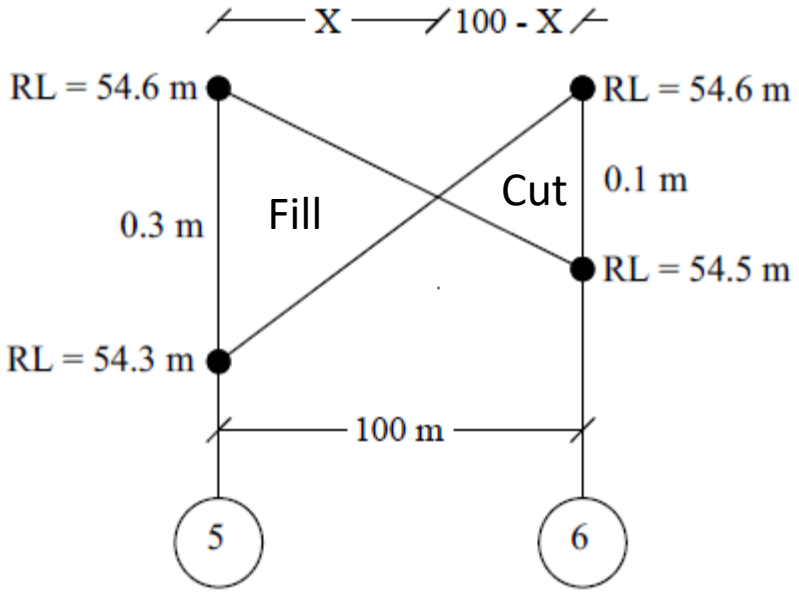






Station	FL (m)	EGL (m)	Depth, d= EGL ~ FL (m)	Average depth, d_{avg} (m)	Area, $A=(b+sd_{avg})d_{avg}$ (m^2)	Length, L (m)	Volume, V = A x L (m^3)	Remark
1	55.0	54.1	0.9					
				1	(8+3 X 1) X 1 = 11	100	1100.00	Fill
2	54.9	53.8	1.1					
				1.2	(8+3 X 1.2) X 1.2 = 13.92	100	1392.00	Fill
3	54.8	53.5	1.3					
				1.25	(8+3 X 1.25) X 1.25 = 14.68	100	1468.00	Fill
4	54.7	53.5	1.2					
				0.75	(8+3 X 0.75) X 0.75 = 7.68	100	768.00	Fill
5	54.6	54.3	0.3					
				0.15	(8+3 X 0.15) X 0.15 = 1.26	75*	95.06	Fill
0	x	x	0					
				0.05	(8+2 X 0.05) X 0.05 = 0.40	25*	10.18	Cut
6	54.5	54.6	0.1					
				0.15	(8+2 X 0.15) X 0.15 = 1.245	100	124.50	Cut
7	54.4	54.6	0.2					
				0.4	(8+2 X 0.4) X 0.4 = 3.52	100	352.00	Cut
8	54.3	54.9	0.6					
				0.45	(8+2 X 0.45) X 0.45 = 4.01	100	400.50	Cut
9	54.2	54.5	0.3					
				0.45	(8+2 X 0.45) X 0.45 = 4.01	100	400.50	Cut
10	54.1	54.7	0.6					
				0.45	(8+2 X 0.45) X 0.45 = 4.01	100	400.50	Cut
11	54.0	54.3	0.3					

Volume of total cutting = 1688.18 m^3
Volume of total filling = 4824.56 m^3



*From figure

$$\frac{X}{0.3} = \frac{100 - X}{0.1} \Rightarrow X = 75'$$

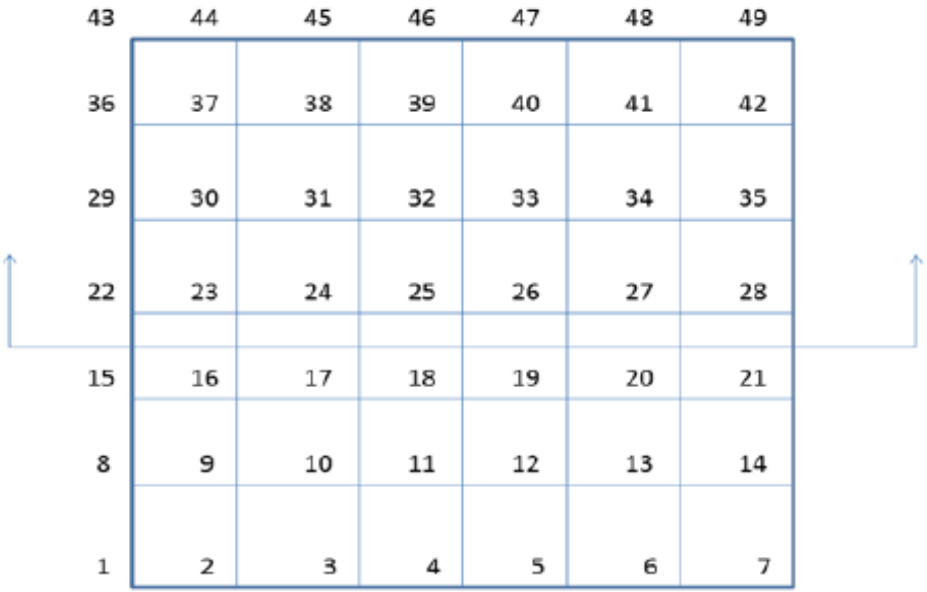
Assignment 1: Calculate the volume of the earthwork required using prismoidal method.

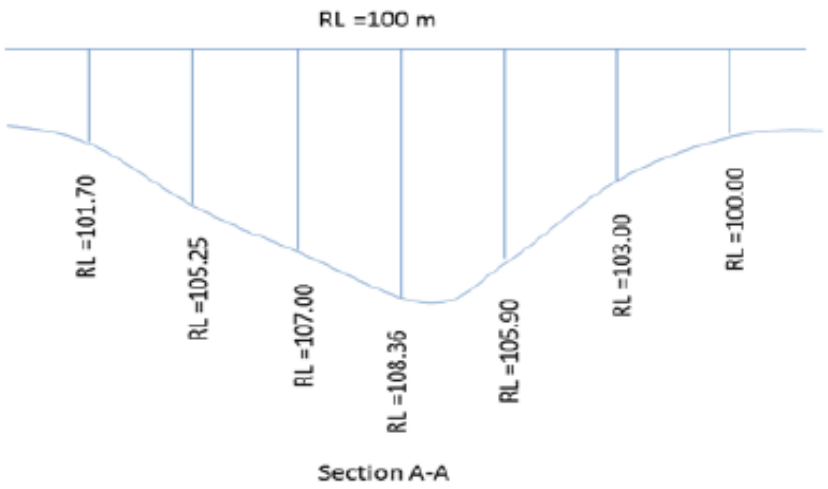
Station	FL(m)	EGL(m)	Depth, d= EGL-FL (m)	Area (A)	Avg. depth, d_{avg} (m)	Area $A_{mid}=(b+sd_{avg})\,d_{avg}$ (m^2)	Length, L (m)	Volume $V=(A_1+4A_{mid}+A_2)\times L/6$ (m^3)	Remarks
1	55	54.1	0.9	A_1					
						A_{1-2}			
2	54.9	53.8	1.1	A_2					
						A_{2-3}			
3	54.8	53.5	1.3	A_3					
						A_{3-4}			
4	54.7	53.5	1.2	A_4					
						A_{4-5}			
5	54.6	54.3	0.3	A_5					
						A_{5-0}			
0	-	-	0	0					
						A_{0-6}			
6	54.5	54.6	0.1	A_6					
						A_{6-7}			
7	54.4	54.6	0.2	A_7					
						A_{7-8}			
8	54.3	54.9	0.6	A_8					
						A_{8-9}			
9	54.2	54.5	0.3	A_9					
						A_{9-10}			
10	54.1	54.7	0.6	A_{10}					
						A_{10-11}			
11	54	54.3	0.3	A_{11}					

Assignment 2: Calculate the volume of the earthwork required using trapezoidal method.

Station	FL(m)	EGL(m)	Depth, d= EGL-FL (m)	Area , A = (b+sd)d	Avg. Area A_{mean} (m^2)	Length, L (m)	Volume $V=A_{mid}xL$ (m^3)	Remarks
1	55	54.1	0.9					
2	54.9	53.8	1.1					
3	54.8	53.5	1.3					
4	54.7	53.5	1.2					
5	54.6	54.3	0.3					
0	-	-	0					
6	54.5	54.6	0.1					
7	54.4	54.6	0.2					
8	54.3	54.9	0.6					
9	54.2	54.5	0.3					
10	54.1	54.7	0.6					
11	54	54.3	0.3					

Assignment 3: Calculate the volume of the earth filling for the ditch. Given RL of filling plane is 100m. Assume each small square to be 50m x 50m. So the total area of the plan is 300m x 300m.





Station	RL (m)	Station	RL (m)	Station	RL (m)	Station	RL (m)	Station	RL (m)
1	101.00	**12**	103.20	**23**	105.25	**34**	104.14	**45**	101.00
2	102.00	**13**	103.90	**24**	107.00	**35**	100.00	**46**	103.20
3	101.70	**14**	101.30	**25**	108.36	**36**	101.10	**47**	101.10
4	102.00	**15**	100.16	**26**	105.90	**37**	104.80	**48**	102.15
5	101.00	**16**	104.15	**27**	103.00	**38**	105.60	**49**	102.00
6	100.60	**17**	106.85	**28**	100.00	**39**	103.00		
7	100.10	**18**	107.65	**29**	101.70	**40**	107.80		
8	100.60	**19**	105.90	**30**	104.80	**41**	104.14		
9	106.00	**20**	104.00	**31**	105.60	**42**	101.00		
10	107.50	**21**	102.00	**32**	103.00	**43**	100.10		
11	105.10	**22**	101.70	**33**	107.8	**44**	102.30		

Hints:

- Calculate depth to be filled at each station
- Calculate volume of each grid to be filled (i.e. $V_{1\text{-}2\text{-}8\text{-}9} = 1/4 \times (d_1 + d_2 + d_8 + d_9) \times A_{1\text{-}2\text{-}8\text{-}9}$)
- Sum up all to get total volume (**Total volume** = $V_{1\text{-}2\text{-}8\text{-}9} + V_{2\text{-}3\text{-}9\text{-}10} + \ldots\ldots\ldots\ldots$)

Hints:

$$V_{1-2-8-9} = \left(\frac{d_1 + d_2 + d_8 + d_9}{4}\right) x A_{1-2-8-9}$$

$$d_1 = 101.00m - 100.00m = 1.00m$$
$$d_2 = 102.00m - 100.00m = 2.00m$$
$$d_8 = 100.60m - 100.00m = 0.60m$$
$$d_9 = 106.00m - 100.00m = 6.00m$$

$$(A_{1-2-8-9}) = \left(\frac{1 + 2 + 0.6 + 6}{4}\right) x50x50 = 6000\ cubic\ meter$$

Material Estimation

Q1. Find out the quantities in 100 cft cement concrete.

Solution- Given, volume of dry concrete= 100 cft (after mixing)
volume of wet concrete= 100 *1.5= 150 cft (before mixing)

Mix ratio (cement: sand: aggregate) = 1:2:4

Cement= 150 x $\frac{1}{1+2+4}$= 21.428 cft= 17.142 bags (1 bag cement= 1.25 cft= 50kg)

Sand= 150 x $\frac{2}{1+2+4}$ = 42. 856 cft

Aggregate= 150 x $\frac{4}{1+2+4}$ =85.712 cft

Material Estimation

Q2. Find out the quantities in 100 cft lime concrete.

Solution- Given, volume of dry concrete= 100 cft (after mixing)
volume of wet concrete= 100 *1.5= 150 cft (before mixing)

Mix ratio (cement: sand: aggregate) = 2:2:7

Cement= 150 x $\frac{2}{2+2+7}$= 27.27 cft

Sand= 150 x $\frac{2}{2+2+7}$= 27.27 cft

Aggregate= 150 x $\frac{7}{2+2+7}$ =95.45 cft

Material Estimation

Q3. Find out the quantities in 100 cft brick masonry. (use a factor of safety of 1.2 for wastage)

Solution- Volume of brickwork= 100 cft
Nominal size of a brick= 10” x 5” x 3” (with mortar)
Actual size of a brick= 9.5” x 4.5” x 2.75” (without mortar)

Number of bricks= $\frac{100\ x\ 1.2}{\frac{10}{12}x\frac{5}{12}x\frac{3}{12}}$= 1382.40= 1383 (say)

Volume of 1383 number of bricks with mortar= 1383 x $\frac{10}{12}$x $\frac{5}{12}$ x $\frac{3}{12}$ =120.052 cft

Volume of 1383 number of bricks without mortar= 1383 x $\frac{9.5}{12}$x $\frac{4.5}{12}$ x $\frac{2.75}{12}$ =94.091 cft

Volume of mortar= (Volume of 1383 number of bricks with mortar)- (Volume of 1383 number of bricks without mortar)

Mortar mix ratio (cement: sand)= 1:4 (say)
Shrinkage factor= 1.5

Cement= 25.965 x 1.5 x $\frac{1}{1+4}$ =7.7895 cft =6.2316 bags (1 bag cement=1.25 cft= 50 kg)

Sand= 25.965 x 1.5 x $\frac{4}{1+4}$= 31.158 cft

Some Important Relations:

1 bag cement= 1.25 cft= 112 lbs= 50 kg
100 cft khoa requires 850 numbers of bricks
200 number of 2.5” nails= 1 seer= 0.9331 kg

1 gallon of plastics emulsion paint/ synthetic enamel paint covers 300 sft of new surface.
1 gallon of distemper covers 250 sft of new surface.
1 tin of snowcem covers 275 square yards.

Brick soling:
Single Brick Flat Soling= 300 number per % sft
Double Brick Flat Soling= 600 number per % sft

Unit Weight of Materials:

Materials	Unit Weight (lb/cft)
Asphalt	66
Bamboo	25
Bitumen	65
Brick Work	120
Brick Chips	80-110
Cement	90
Coal (loose)	50
Earth (compacted and damp)	100
Earth (dry and loose)	90
Lime Concrete	120
Plain Concrete	144
Reinforced Cement Concrete	150
Mild Steel	490
Cast Iron	450

Materials	Unit Weight (lb/cft)
Wrought Iron	480
Pebbles	90
Sand (dry to wet)	100-120
Clay (dry to wet)	115-130
Shingles	90
Gravel	90
Surkhi	80
Surkhi Mortar	85-120
Tar	90
Garjan Timber	45-50
Sal Wood	60-65
Sundari Wood	55-65
Teak Wood	45-55
Water at $4^o c$	62.45

Practice Problems:

Q1. Find out the ingredients required for making 20’ x 18’ concrete slab (thickness=5”). Given, mix ratio= 1:1.5:3.
Answer- cement= 32.72 bags; sand (fine aggregate)= 61.36 cft; brick/ stone chips (coarse aggregate0= 122.73 cft

Q2. Suppose, you have 50 bags of cement. What volume of concrete you can make from this 50 bags cement with a mix ratio of 1: 1.25: 2.5.
Answer- 197.92 cft dry concrete

Q3. Suppose you have 5000 Nos. of full size brick. You want to make a brick wall. What will be the volume of brick work?
Answer- 434.03 cft

Q4. In question 3, determine the number of cement bags and volume of sand if mortar mix ratio is 1:4.
Answer- 22.53 bags cement; 112.63 cft sand.

Q5. Given a 10” thick brick wall having a length of 30’ and height of 10’. The wall contains two doors (7’ high and 3.5’ wide) and two windows (4’ high and 6’ wide) and a continuous lintel (25’ long and 6” high). Using a factor of safety of 1.2 and cement: sand ratio in mortar= 1:4, Estimate the number of bricks, number of bags of cement an d sand required for the construction of the wall.
Answer- number of bricks= 2195; cement= 9.89 bags; sand= 49.43 cft.

Practice Problems:

Q6. Lime concrete (3” thick) was cast over an RCC roof having a plan area of 50’ x 30’. Determine the ingredients (mix ratio=2:2:7).
Answer- Lime= 102.27 cft; surkhi= 102.27 cft; brick chips= 357.95 cft

Q7. Find out the ingredients required for the RCC foundation up to gorund level. Given mix Ratio= 1:2:4.
Answer- cement= 23.72 bags; sand= 59.298 cft; brick/ stone chips= 118.597 cft

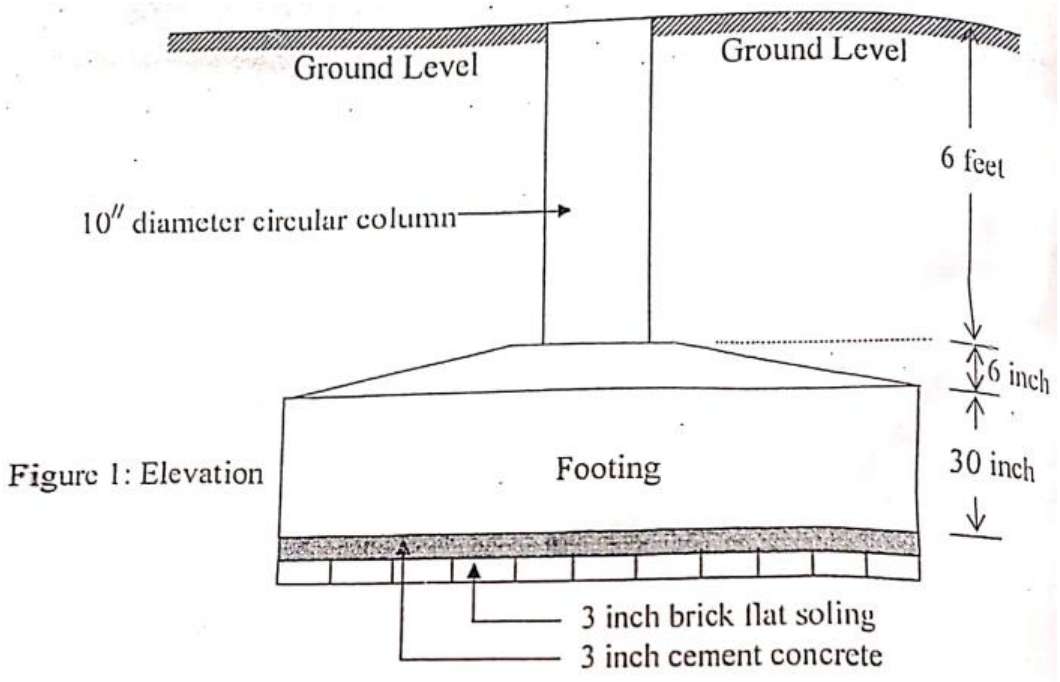


Figure 1: Elevation

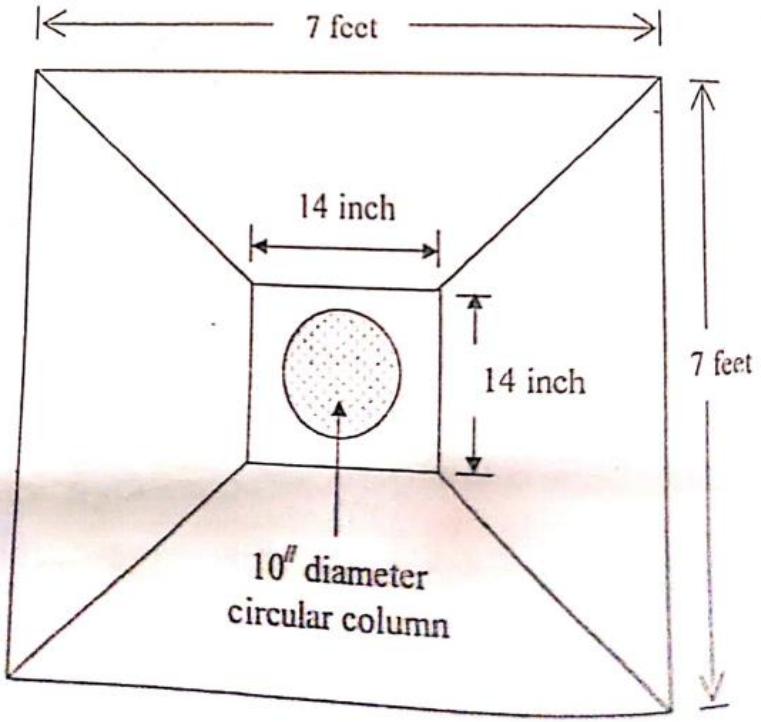


Figure 2: Plan